

EDA: Bivariate & Multivariate Analysis

Module 21 Study Notes • Exploratory Data Analysis

1. Moving Beyond a Single Variable

While Univariate Analysis looks at one column in isolation, **Bivariate Analysis** examines the relationship between exactly two columns. **Multivariate Analysis** extends this to three or more columns simultaneously. The choice of visualization technique depends entirely on the data types being compared.

2. Numerical vs. Numerical Data

When investigating the relationship between two continuous numerical variables:

A. Scatter Plot

The standard tool to see if a linear or non-linear correlation exists between an X and Y variable.

Adding Multivariate Context (Seaborn): You can pack up to 5 variables into a single scatter plot using Seaborn parameters:

- **hue** : Colors the dots based on a categorical variable (e.g., Male vs. Female).
- **style** : Changes the marker shape based on another category (e.g., Dot for Smoker, Cross for Non-Smoker).
- **size** : Adjusts the marker size based on a numerical or categorical value (e.g., Size of the party).

B. Line Plot

A specialized form of scatter plot used **strictly when the X-axis represents Time** (e.g., years, months, dates). It connects data points chronologically to display trends.

3. Numerical vs. Categorical Data

When comparing a continuous numerical variable against distinct categories:

A. Bar Plot

By default, Seaborn's bar plot calculates the **average (mean)** of the numerical variable for each category. (e.g., Average ticket fare paid by Class 1, Class 2, and Class 3 passengers). Adding a **hue** parameter further splits the bars (e.g., Average fare by Class, split by Gender).

B. Side-by-Side Boxplots

Plotting multiple boxplots next to each other allows you to compare the 5-number summary (spread, median, and outliers) of a numerical variable across different categories.

C. Overlaying KDEs (Distplots)

You can plot multiple Kernel Density Estimation (KDE) curves on the same graph to see how probability distributions overlap. For example, overlaying the age distribution of "Survivors" and "Non-Survivors" can instantly reveal which age groups were most likely to survive.

4. Categorical vs. Categorical Data

When comparing two discrete categories against each other:

A. Heatmap

First, use **pd.crosstab()** or a **pivot_table** to create a matrix counting the occurrences of each combination. Then, pass this matrix to a **Heatmap**. Heatmaps use color intensity to quickly highlight areas of high density or frequency.

B. Clustermap

A variation of a heatmap that incorporates **Hierarchical Clustering** (Dendrograms). It physically moves similar rows and columns closer together, instantly revealing groups or clusters of similar behavior in your data.

5. The High-Level Overview: Pairplot

When you have a dataset with many numerical columns, plotting individual scatter plots is tedious. Using **sns.pairplot(df)** automatically generates a grid displaying a scatter plot for *every possible combination* of numerical columns in your dataset. It is the fastest way to get a bird's-eye view of all numerical relationships.